

Probability and Statistics (MT-2005)

Homework #01 (d)

Visualization of Bivariate Data

Problem Statement

Given a dataset with two variables, create a visualization that effectively represents the relationship between them.

Solution

Solution

A scatter plot is an effective way to visualize the relationship between two quantitative variables. Below is an example of Python code using Matplotlib to generate a scatter plot:

```
import matplotlib.pyplot as plt
import numpy as np

# Sample Data
x = np.random.rand(30) * 10 # Independent variable
y = 2 * x + np.random.randn(30) * 2 # Dependent variable with noise

plt.scatter(x, y, color='blue', alpha=0.7)
plt.xlabel('Independent Variable')
plt.ylabel('Dependent Variable')
plt.title('Scatter Plot of Bivariate Data')
plt.grid(True)
plt.show()
```

This scatter plot visually demonstrates any correlation or pattern between the two variables.